

LeetCode 661: Image Smoother

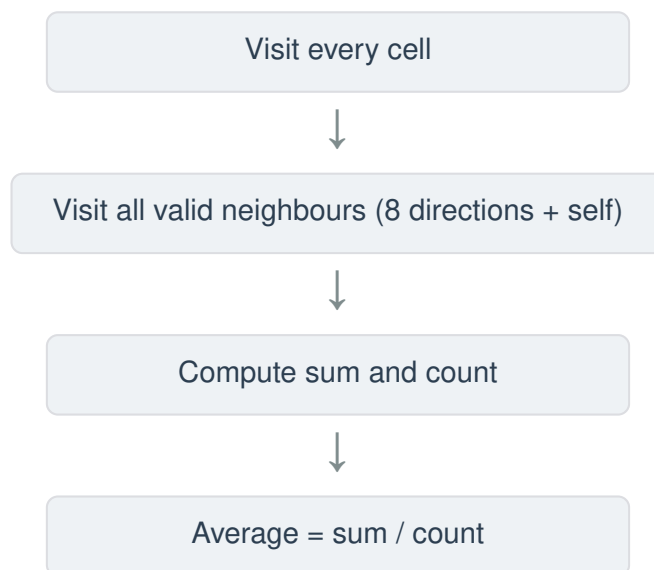
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Analysis & Notes

Because each cell has at most **9 neighbours** (including itself), the brute force approach evaluated per cell requires checking a maximum of 9 adjacent cells. Since this is a fixed upper bound, the per-cell computation is completed in **constant time $O(1)$** .

Consequently, for a matrix of size $m \times n$, the overall time complexity gracefully scales to **$O(m * n)$** .

Execution Pattern



C++ Implementation

```
class Solution {  
public:
```

```

vector<vector<int>> imageSmoother(vector<vector<int>>& img)
{
    int n=img.size();
    int m=img[0].size();

    vector<vector<int>> ans(n,vector<int>(m));

    vector<pair<int,int>> dir={{-1,-1},{-1,0},{-1,1},{1,-1},
{1,0},{1,1},{0,-1},{0,0},{0,1}};

    for(int i=0;i<n;i++){
        for(int j=0;j<m;j++){
            int sum=0;
            int cnt=0;

            for(auto d:dir){
                int nr=i+d.first;
                int nc=j+d.second;

                if(nr>=0 && nr<n && nc>=0 && nc<m){
                    sum+=img[nr][nc];
                    cnt++;
                }
            }

            ans[i][j]=sum/cnt;
        }
    }

    return ans;
}
};

```